



Cannabis use, other substance use, and co-occurring mental health concerns among youth presenting for substance use treatment services: Sex and age differences

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ABSTRACT

Cannabis use among youth is of significant concern given the potential negative health impacts on young people. This study describes the mental health characteristics, cannabis use, and substance use profiles of youth aged 14 to 24 years seen at the Youth Addiction and Concurrent Disorders Service at the Centre for Addiction and Mental Health, through an age and gender lens. Almost all participating youth indicated lifetime cannabis use. Substance use and mental health profiles among those indicating a history of cannabis use differed along sex and age. Male youth were more likely to use cannabis more frequently, while female youth were more likely to use greater variety of substances and display more co-occurring mental health problems. Adolescent problematic substance use tended to co-occur with externalizing problems whereas problematic young adult substance use tended to co-occur with internalizing problems. Implications for the integrated, coordinated care of substance use and mental health concerns, with attention to the unique needs of subgroups of youth, are discussed.

1. Introduction

Cannabis is the illegal drug that is the most widely used around the world (United Nations Office on Drugs and Crime, 2016), and its use among youth is common especially in Canada (World Health Organization & Regional Office for Europe, 2012). Canadian youth ranked first in the world for cannabis use in an international survey, with 33% of youth indicating lifetime cannabis use by the age of 15, and 18% indicating having used cannabis within the past 30 days (World Health Organization & Regional Office for Europe, 2012). Data from the Canadian Community Health Survey shows that among Canadian youth between the ages of 15 and 24, some 17% to 30% have used cannabis, and 2% to 5% use it on a daily basis (Rotermann & Langlois, 2015). Among young Canadians seeking substance abuse services, cannabis has been reported to be the most common substance of concern (Urbanoski, Strike, & Rush, 2005).

However, due to the rapid changes in brain development during adolescence, young people show marked vulnerability to the impacts of substance abuse, including cannabis (Bava & Tapert, 2010). Notably, cannabis use among youth, especially high-frequency use (i.e., daily or near daily use), has been associated with a variety of negative cognitive, physical, and mental health consequences, some of which could

lead to long-term deficits. For example, high-frequency cannabis during adolescence has been linked to attention, learning, memory and executive function problems (Broyd, van Hell, Beale, Yücel, & Solowij, 2016); reduced academic achievement (Silins et al., 2014); respiratory problems (Brook, Stimmel, Zhang, & Brook, 2008); and long-term cancer risk (Callaghan, Allebeck, & Sidorchuk, 2013).

High-frequency cannabis use has also been associated with a variety of mental health and substance use problems. There has been some research suggesting a link between adolescent cannabis use and depression, anxiety, externalizing disorders, and suicidality (e.g., Horwood et al., 2012; Monshouwer et al., 2006; Silins et al., 2014), although the causality and directionality of the relationships have not been definitively determined. Relationships have been found between high-frequency cannabis use and psychotic experiences, especially among those who are predisposed to developing a psychotic disorder (Radhakrishnan, Wilkinson, & D'Souza, 2014). Studies have also shown that early, high frequency cannabis use is associated with the use of other 'harder' illicit drugs such as opiates and cocaine during adolescence and emerging adulthood, as well as later in life, although the majority of cannabis users do not necessarily progress to using other illicit drugs (Fergusson & Horwood, 2000; Lynskey et al., 2003).

In the context of high rates of cannabis use among young people in

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Canada, there is a strong public perception that cannabis is, to a large extent, a harmless drug. A focus group study with 76 youth sampled across Canada found that many adolescents between the ages of 14 and 19 perceived cannabis to be safe, natural, and non-addictive, and many of the youth were unaware, indifferent, or unsure about the negative consequences associated with cannabis use (Porath-Waller, Brown, Frigon, & Clark, 2013). The public perception of harmlessness, despite potential negative impacts on the developing brain, is of particular concern in Canada due to the impending legalization of non-medical cannabis use (Health Canada, 2016). Indeed, in the pre-legalization era, cannabis is readily available.

Some aspects of cannabis use and youth remain understudied and require attention as Canada moves forward toward legalization. Notably, until recently female substance use and sex/gender differences in cannabis use had been understudied. It has been demonstrated that more males than females use cannabis and meet the criteria for cannabis abuse and dependence (Leatherdale & Burkhalter, 2012; Pearson, Janz, & Ali, 2013). Because males are more likely to use cannabis, they represent the majority of the participants in much of the research conducted about youth cannabis use. However, it has been suggested that early cannabis initiation may interact with aspects of health and functioning differently across genders. For example, girls may transition from cannabis initiation to regular use more quickly (Schepis et al., 2011), have poorer socioeconomic and mental health status than boys who use cannabis (Tu, Ratner, & Johnson, 2008), and drop out of school earlier (van Ours & Williams, 2009). The potential gender differences highlight the importance of further examining cannabis use profiles of youth and young adults using a gender lens. By examining these differences among youth and young adults who are seeking treatment, it may be possible to shed light on treatment issues that will enable services to better meet the needs of young people. Similarly, there is emerging discussion regarding the differential needs of adolescents and transition-aged youth presenting for substance use treatment (e.g., Bowers, Cleverley, Di Clemente, & Henderson, 2017; Bukstein, 2017; Ilana, Joanna, Michele, & Abby, 2015), which also warrant further study.

Given the urgent need to understand cannabis use among young people in the context of impending legalization and the paucity of research into the relationship between sex, age and cannabis use in treatment-seeking youth in Canada, the objective of the current study is to explore the cannabis use profiles among male and female youth and young adults in a hospital-based outpatient substance use treatment-seeking sample in Canada. More specifically, this study describes the rates of cannabis use, substance use profiles, and mental health characteristics of male and female adolescents and young adults seeking treatment in a Canadian, hospital-based, youth addiction and concurrent disorders service with a view of informing future service and research directions.

2. Methods

2.1. Sample characteristics

The sample consisted of 1378 youth receiving services at the Youth Addiction and Concurrent Disorder Service at the Centre for Addiction and Mental Health, a large urban mental health hospital in Toronto, Canada. This is an outpatient treatment service for youth 14 to 24 years of age offering treatment for substance use with or without co-occurring mental health concerns. Since the clinic accepts referrals from multiple routes, ranging from physician referral to self-referral, no diagnosis is required to receive services. All youth who attended an initial orientation session introducing them to the services between 2008 and 2016 were eligible for the research study. A total of 1595 youth were assessed and approached for research consent, and 86% ($n = 1378$) provided consent for the use of their data for research purposes.

2.2. Measures

At the orientation session, participants were asked to fill out a battery of questionnaires, including a demographic information form and questionnaires related to their substance use and mental health needs.

2.2.1. Demographic information form

The demographic information form included questions about participant's age at orientation, sex, race/ethnicity, country of origin, first language, employment, education, financial support, living arrangements and legal involvement.

2.2.2. Adolescent Alcohol and Drug Involvement Scale (AADIS)

The Adolescent Alcohol and Drug Involvement Questionnaire (Moberg, 2005) asks youth to rate whether they have ever used, and how frequently they have used, different types of substances using an 8-point scale ranging from never used (0) to several times a day (7). The substances included were: alcohol, tobacco, cannabis, hallucinogens, amphetamines, powder cocaine, rock cocaine, barbiturates, opiates, benzodiazepines, and "other drugs." Beginning in September 2013, for each of the substances the youth had used at least once, they were also asked to indicate the age of first use, resulting in a subsample of the youth ($n = 390$) with age-of-first-use data. Results are presented for the substances most commonly endorsed.

2.2.3. Target substance

Using an open-ended question, youth were asked which substance they found to be most problematic, of greatest concern, or wanted to change (either to reduce or to abstain from). In response to this question, youth could indicate as many substances as they wished as their target substances.

2.2.4. Global Appraisal of Individual Needs – Short Screener (GAIN-SS)

The GAIN Short Screener (GAIN-SS; Chestnut Health Systems, 2010) is a short self-report screening tool based on the Global Appraisal of Individual Needs – Initial (Dennis, 2003; Dennis, White, Titus, & Unsicker, 2008). With 20 items, the GAIN-SS quickly identifies youth who are likely to meet criteria for a diagnosis of an internalizing disorder (e.g., depression, anxiety), externalizing disorder (e.g., ADHD, conduct problems), or substance use disorder, as well as identifying significant crime and violence problems. The youth are asked to report on symptoms using temporal response options: Never (0), More than a year ago (1), 2 to 12 months ago (2), and Past month (3). Due to a change in the measurement protocol, this was expanded into Never (0), More than a year ago (1), 4 to 12 months ago (2), 2 to 3 months ago (3), and Past month (4); however, these ratings were collapsed into the original 0–3 point scale for the purposes of the analyses. The number of past-year symptoms (items with a score of 2 or 3) is counted, and if the count of items with past-year symptoms is 3 or greater for any subscreener, the youth is considered as having a high probability of a diagnosis, in accordance with the norms for the scale. GAIN-SS has good internal consistency (Cronbach's $\alpha = 0.96$ for the total screener), sensitivity (90% or greater), and specificity (92% or greater), and shows a high correlation ($r = 0.84$ to 0.94) with the original GAIN-I domains (Dennis, Chan, & Funk, 2006).

In addition to the 20 standard items of the GAIN-SS, seven extension items were added to the screener with the permission of Chestnut Health Systems in 2006, based on community/service provider feedback by co-author JH and team. These items were designed to be used as indicators suggestive of areas for further assessment and reflect domains that are covered in GAIN-I but not in the original GAIN-SS, including eating-concerns, traumatic stress, distorted thinking, overuse of videogames and internet, and gambling concerns. Although these extension items are not part of the original GAIN-SS, a cross-sectoral study has demonstrated their utility (Henderson, Chaim, Hawke, & National

Youth Screening Project, 2017). The response options for the extension items were the same as for those of the standard items, and the same past-year cut-offs were also used.

2.3. Statistical analysis

The main independent variables used in this study were sex (male vs. female) and age (adolescents: 14–17, young adults: 18–24). For continuous outcome variables, *t*-tests were performed, and for dichotomous variables, chi-square tests were used to examine group differences. All statistical tests were analyzed using IBM SPSS Statistics for Windows, Version 24.0 (IBM Corp., 2016). To correct for multiple hypothesis testing, the Bonferroni method was used within sets of analyses. To examine the extent of overlap among internalizing, externalizing, and substance use problems among youth using cannabis, area-proportional Venn diagrams were drawn using the EulerAPE software (Micallef & Rodgers, 2014).

3. Results

3.1. Demographic information

The sample consisted of 1378 youth (867 male, 486 female, 8 trans/other, 17 missing)² who provided consent to participate in the research study between 2008 and 2016. In all, 96% of participants presenting for treatment indicated lifetime cannabis use. There were no sex differences in whether the youth had tried cannabis or not, (sex: $\chi^2(1, N = 1289) = 2.08, p = 0.15$).

The remaining analyses compare cannabis users, by sex and age (Table 1). All youth were between the ages of 14 and 24 years, with a mean age among cannabis-using youth of 19.64 years old ($SD = 2.48$). There were no significant differences in age between male and female participants, $t(1210) = 0.02, p = 0.98$. Eighty-seven percent of the youth were born in Canada, 68% identified as White/European descent; and 91% spoke English as their first language, which did not differ statistically between boys/young men and girls/young women.

There were some statistically significant sex differences and statistical trends observed in employment/student status, living arrangements, and legal involvement. Compared to female youth, male youth had a higher probability of being “NEET,” i.e., not engaged in education, employment or training, $\chi^2(1, N = 1126) = 5.22, p = 0.02$. In addition, significantly more boys/young men than girls/young women were living with their parents or other family members, $\chi^2(2, N = 1110) = 16.15, p < 0.001$, and more than half (55%) of the boys/young men and one-third (35%) of the girls/young women had legal system involvement at some point in their lives, $\chi^2(1, N = 855) = 28.34, p < 0.001$.

3.2. Analysis of sex differences

3.2.1. Patterns of Cannabis and other substance use

There was a significant difference between male and female youth in the frequency of cannabis use. Boys/young men were more likely than girls/young women to use cannabis several times a week or more, ($\chi^2(1, N = 1231) = 14.37, p < 0.001$).

On the other hand, girls/young women indicated having used a greater variety of substances than boys/young men, $t(1175) = 4.60, p < 0.001$. On average, male participants indicated that they had tried 5.6 ($SD = 2.2$) different types of substances and female participants indicated that they tried 6.2 ($SD = 2.2$) different substances.

Sixty-eight percent of the boys/young men and 55% of the girls/

Table 1

Demographic characteristics of male vs. female cannabis-using youth.

Variable name	Male		Female		<i>t</i>	<i>p</i>
	(N = 786)		(N = 446)			
	<i>M</i>	(<i>SD</i>)	<i>M</i>	(<i>SD</i>)		
Age at Intake (in years)	19.63	2.50	19.63	2.44	0.02	0.98
	<i>n</i>	(%)	<i>n</i>	(%)	χ^2	<i>p</i>
White/European	493	66%	302	71%	3.52	0.06
Born in Canada	637	86%	366	88%	0.83	0.36
English as a first language	666	90%	387	93%	2.60	0.11
Not Engaged in Education, Employment or Training (NEET)	273	38%	128	31%	5.22	0.02
Government support (welfare, pension, employment insurance, family benefits)	176	24%	110	26%	0.67	0.41
Living arrangements						
On own/shared living	127	18%	107	27%	16.15	< 0.001 ^a
Living with parent(s)/ family	546	76%	254	65%		
Precarious living arrangements	46	6%	30	8%		
Legal system involvement	312	55%	101	35%	28.34	< 0.001

^a *df* = 2; all other chi-square analyses *df* = 1; sample sizes vary due to missing data.

young women who had used cannabis indicated that cannabis was one of their target substances for change or one of their problematic substances. Male participants were more likely than female participants to specify cannabis as a target substance, $\chi^2(1, N = 1110) = 18.64, p < 0.001$, even when controlling for the number of types of substances used, Mantel-Haenszel $\chi^2(1, N = 1069) = 8.21, p = 0.004$, and frequency of use, Mantel-Haenszel $\chi^2(1, N = 1109) = 9.54, p = 0.002$.

Fig. 1 shows the average age of onset for cannabis and other substances, subdivided by sex. The average age of onset for cannabis use across the sexes was 14.4 years ($SD = 2.1$). Among male participants, the average age was 14.5 years ($SD = 2.2$), and among female participants, 14.3 years ($SD = 1.9$); the sex difference was not statistically significant, $t(313) = 0.82, p = 0.41$. Sex differences were not significant for the age of onset of any of the individual substances at a Bonferroni-corrected *p* value of 0.05/7 = 0.007, although trends toward significance were observed, with female participants seeming to begin using the substance earlier for tobacco ($p = 0.05$), alcohol ($p = 0.01$) and opiates ($p = 0.04$).

Among young people who used both cannabis and tobacco, age of onset was not significantly different, at 14.3 ($SD = 2.0$) for cannabis, compared to 14.6 ($SD = 2.2$) for tobacco, $t(295) = 2.52, p = 0.012$ (Bonferroni-corrected *p*-value = 0.05/6 = 0.008). Among participants who used cannabis as well as alcohol, hallucinogens, amphetamines, powder cocaine, and/or opiates, significant differences were found in the age of onset of use for each substance pair. The average age of onset of cannabis use was significantly older than that of alcohol, but significantly younger than all other substances. For cannabis and alcohol use, the average age was 14.4 ($SD = 2.0$) for cannabis, and younger for alcohol, at 13.8 years ($SD = 2.2$), $t(291) = 4.37, p < 0.001$. For users of cannabis and other illicit drugs (hallucinogens, amphetamines, powder cocaine, opiates), a similar profile emerged across substances, with average cannabis onset ages at approximately 14 and average ages of onset of the other substance at over 16 (all *p* values < 0.001).

3.2.2. GAIN-SS

Among all participants who had lifetime cannabis use, mental health status was explored by sex (Table 2). There were sex differences in 3 out of the 4 standard domains assessed on the GAIN-SS, at a Bonferroni-corrected *p* value of 0.05/11 = 0.0045 for the GAIN-SS as a

² There were 3 youth who identified as trans, and 5 as other. Due to the small sample size, these 8 youth were excluded from the sex-based analysis, but included in other analyses.

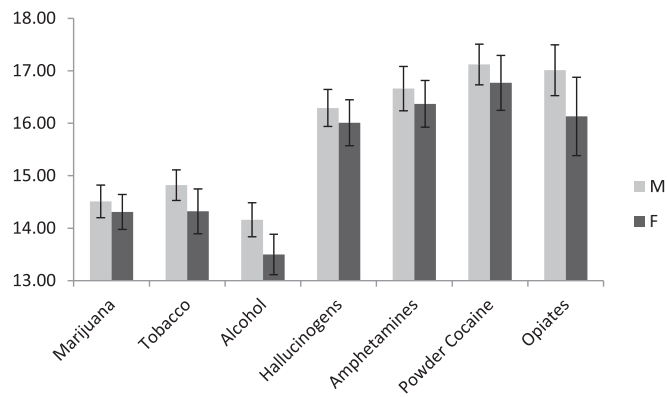


Fig. 1. Age of first use by substance type and sex with 95% confidence intervals.

disorder ($\chi^2[1, N = 1218] = 14.39, p < 0.001$), and substance use disorder ($\chi^2[1, N = 1217] = 11.61, p = 0.001$), although rates of endorsement were high for both male and female participants for both screeners. There were no statistically significant sex differences in the likelihood of screening positively for crime or violence problems ($\chi^2[1, N = 1215] = 0.455, p = 0.50$).

There were statistically significant sex differences in 4 out of the 7 GAIN-SS extension items. Eating-related concerns were more frequent among female participants: more than half of the girls/young women reported skipping meals or purging, versus about a quarter of the boys/young men: $\chi^2(1, N = 1212) = 126.65, p < 0.001$; and 60% of female participants reported binge eating compared to slightly over a quarter of the male participants, $\chi^2(1, N = 1211) = 121.40, p < 0.001$. Eighty percent of female participants had disturbing memories and dreams, versus 59% of male participants: $\chi^2(1, N = 1206) = 58.67, p < 0.001$. However, paranoia-related thought disturbances were significantly

Table 2

Cannabis use and mental health profiles of male and female cannabis users in treatment setting.

	Male		Female		χ^2/t	p
	<i>n</i>	%	<i>n</i>	%		
Substance use profiles						
Lifetime cannabis use - <i>n</i> (%)	786	96%	446	94%	2.08	0.15
Very high frequency cannabis use: daily/several times a day – <i>n</i> (%)	410	52%	206	46%	4.15	0.04
High frequency cannabis use: several times a week or more – <i>n</i> (%)	545	69%	262	59%	14.37	< 0.001
Cannabis as target substance – <i>n</i> (%)	478	68%	220	55%	18.64	< 0.001
Age at first cannabis use - <i>m</i> (sd)	14.5	(2.2)	14.3	(1.9)	0.82	0.41
Number of substances used in lifetime – <i>m</i> (sd)	5.6	(2.2)	6.2	(2.2)	4.60	< 0.001
GAIN-SS Subscreeener Endorsement – <i>n</i> (%)						
Internalizing disorder	588	76%	421	95%	72.82	< 0.001
Externalizing disorder	532	69%	350	79%	14.39	< 0.001
Substance use disorder	644	83%	400	90%	11.61	0.001
Crime/violence problems	231	30%	124	28%	0.45	0.50
GAIN-SS extension items endorsement – <i>n</i> (%)						
Skipping meals, purging	181	23%	245	56%	126.65	< 0.001
Binge eating	216	28%	265	60%	121.40	< 0.001
Disturbed by memories/dreams	451	59%	353	80%	58.67	< 0.001
Had thoughts that people are watching/following/out to get you	339	44%	230	52%	6.98	0.008
Saw/heard things no one else could	181	25%	116	28%	1.15	0.28
Problematic videogame playing/internet use	222	31%	104	26%	2.40	0.12
Problematic gambling	56	8%	20	5%	3.02	0.08

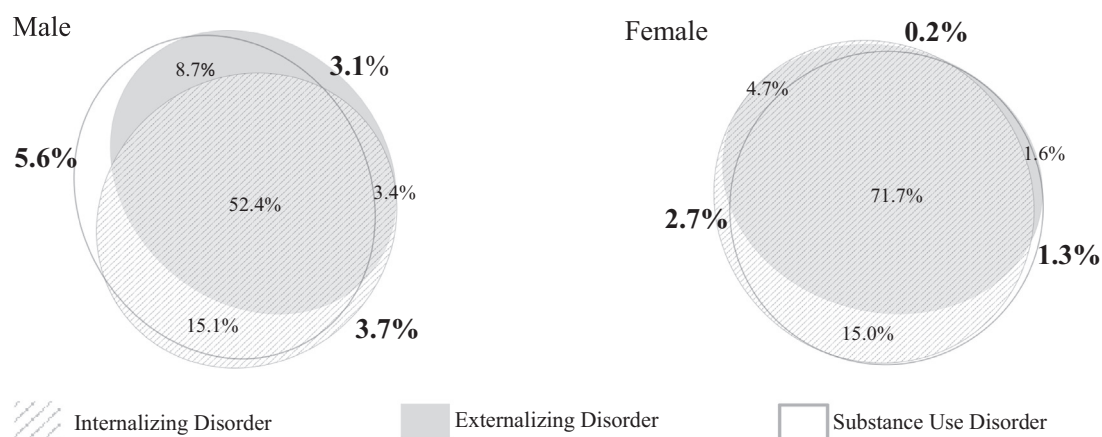


Fig. 2. Concurrent mental health and substance use problems among male and female cannabis users seeking treatment.

Note: 6.4% of male participants and 1.8% of female participants did not screen positive for any of the three categories of disorders. Bold indicates the non-concurrent cases.

whole. Almost all the female youth and roughly three-quarters of the male youth screened positively for an internalizing disorder, $\chi^2(1, N = 1218) = 72.82, p < 0.001$. Girls/young women were also more likely than boys/young men to screen positively for an externalizing

more frequent among girls/young women (52%) compared to boys/young men (44%); $\chi^2(1, N = 1201) = 6.98, p = 0.008$. There were no significant sex differences hallucination-like symptoms, problem videogame playing, or problem gambling.

Fig. 2 shows the extent of concurrent substance use, internalizing, and externalizing disorders among male and female cannabis-using youth in a treatment-seeking sample. Seventy-two percent of female and 52% of the male cannabis users had a high probability of having all three categories of disorders. In fact, substance use disorders, internalizing disorders, or externalizing disorders occurring independently without any co-occurring conditions were rare in this sample of youth seeking substance use treatment: only 6% of male participants and 1% of female participants screened positively for substance use disorder without any mental health concerns. Among cannabis-using participants who screened positive for problematic substance use, internalizing, and externalizing disorders together, 72% of male participants and 61% of female participants reported using cannabis daily or near daily, a statistically significant difference, $\chi^2(1, N = 732) = 10.24$, $p = 0.001$.

3.3. Analysis of age group differences

To examine the effect of age on cannabis use and mental health outcomes, cannabis-using participants were dichotomized into two age groups: adolescents (14–17 years; $n = 274$) and young adults (18–24 years; $n = 956$). The average age of the adolescents and the young adults was 16.23 ($SD = 0.84$) and 20.61 ($SD = 1.85$) respectively. There was no significant difference in the distribution of male and female participants across the two age groups, $\chi^2(1, N = 1212) = 0.35$, $p = 0.85$.

3.3.1. Patterns of cannabis use

Table 3 summarizes the cannabis use patterns by age group. Both adolescents (98%) and young adults (95%) in our treatment-seeking sample almost unanimously indicated having used cannabis at some point in their lives. Adolescents presenting for treatment started using cannabis roughly one year earlier than youth who were young adults at the time of treatment presentation (13.6 vs 14.7 years), $t(288) = 3.44$, $p = 0.001$. More than half (57%) of the adolescents indicated using cannabis daily or several times a day (“very high frequency use”), compared to 47% of young adults. Three-quarters of the adolescents indicated using cannabis several times a week or more (“high frequency use”), compared to 62% of young adults. Adolescents were more likely than the young adults to indicate very high ($\chi^2[1, N = 1229] = 7.92$, $p < 0.005$) and high ($\chi^2[1, N = 1229] = 16.11$, $p < 0.001$) frequency cannabis use. Adolescents were also more likely than young adults to

indicate cannabis use as one of their substances of concern or treatment target, $\chi^2(1, N = 1107) = 18.87$, $p < 0.001$, even after controlling for frequency of cannabis use, Mantel-Haenszel $\chi^2(1, N = 1064) = 8.31$, $p = 0.004$, and number of substance used, Mantel-Haenszel $\chi^2(1, N = 1064) = 12.40$, $p < 0.001$.

3.3.2. GAIN-SS

Table 3 summarizes the age group differences on the GAIN-SS standard domains and extension items among cannabis-using adolescents vs. young adults. Results of the standard domains show that young adults who have used cannabis at some point in their lives had a higher probability of screening positive for an internalizing disorder than adolescents, $\chi^2(1, N = 1216) = 29.99$, $p < 0.001$, but adolescents had a higher probability of screening positive for an externalizing disorder, $\chi^2(1, N = 1216) = 17.94$, $p < 0.001$, or having issues related to crime and violence, $\chi^2(1, N = 1213) = 25.67$, $p < 0.001$. There was no significant age group difference in the substance use domain. The only statistically significant age group difference found for the extension items was regarding disturbing memories and dreams, with young adults indicating a higher probability of recent problems than adolescents, $\chi^2(1, N = 1204) = 13.31$, $p < 0.001$.

Fig. 3 shows the overlap among substance use disorders, internalizing disorders, and externalizing disorders according to the past year symptoms on the GAIN-SS among cannabis-using adolescents and young adults. Results show concurrent disorders were nearly universal and that roughly 60% of both adolescents and young adults had a high probability of having all 3 categories of disorders. The main difference between adolescents and young adults is that substance use disorders were more likely to co-occur with externalizing disorders in adolescents (12.0%), and with internalizing disorders in young adults (17.5%). Among those who met screening criteria for substance, internalizing, and externalizing three-way concurrent disorders, the majority reported using cannabis near daily or more, with significantly higher rates for adolescents (77%) than young adults (64%), $\chi^2(1, N = 733) = 10.84$, $p = 0.001$.

4. Discussion

Among adolescents and young adults seeking services for substance use disorders, our findings show that lifetime cannabis use was almost universal, and the youth initiated cannabis use earlier than the Canadian average (Canadian Centre on Substance Use and Addiction,

Table 3
Cannabis use and mental health profiles of adolescents and young adults in treatment setting.

	Adolescents ^a		Young adults ^a		χ^2/t	p
Substance use profiles						
Lifetime cannabis use - n (%)	274	98%	956	95%	5.85	0.02
Very high frequency cannabis use: daily/several times a day – n (%)	156	57%	454	47%	7.92	0.005
High frequency cannabis use: several times a week or more– n (%)	206	75%	596	62%	16.11	< 0.001
Cannabis as target substance – n (%)	174	75%	524	60%	18.87	< 0.001
Age at first cannabis use - m (sd)	13.6	(1.3)	14.7	(2.2)	3.44	0.001
Number of substances used in lifetime – m (sd)	5.4	(2.3)	5.9	(2.2)	3.50	< 0.001
GAIN-SS subscreener endorsement – n (%)						
Internalizing disorder	195	72%	814	86%	29.99	< 0.001
Externalizing disorder	224	83%	658	70%	17.94	< 0.001
Substance use disorder	223	82%	818	87%	3.27	0.07
Crime/violence problems	112	41%	240	25%	25.67	< 0.001
GAIN-SS extension items endorsement – n (%)						
Skipping meals, purging	86	32%	338	36%	1.32	0.25
Binge eating	90	34%	390	41%	5.39	0.02
Disturbed by memories/dreams	154	58%	652	70%	13.31	< 0.001
Had thoughts that people are watching/following/out to get you	130	49%	443	47%	0.23	0.63
Saw/heard things no one else could	80	31%	217	24%	5.35	0.02
Problematic videogame playing/internet use	89	35%	236	27%	4.96	0.03
Problematic gambling	18	7%	58	7%	0.02	0.88

^a Adolescents: age 14–17 years, Young adults: age 18–24 years.

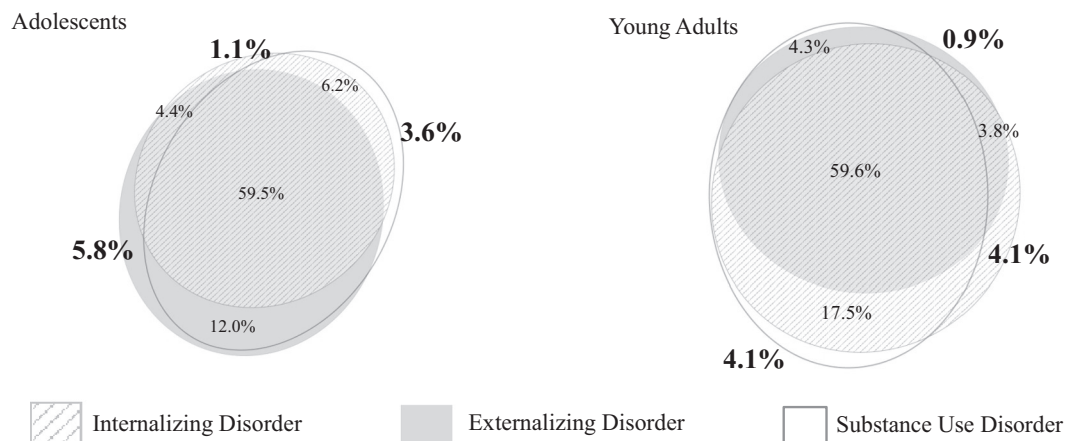


Fig. 3. Concurrent mental health and substance use problems among adolescents and young adults.

Note. 6.2% and 4.3% of adolescents and young adults respectively did not screen positive for any of the three categories of disorders. Bold indicates the non-concurrent cases.

2017; Government of Canada, 2017; Health Canada, 2012). Young men were more likely to use cannabis more frequently and identify cannabis as their treatment target more often than did young women, while young women were more likely than young men to use a greater number of different substances and display three-way concurrent problems across the internalizing, externalizing, and substance use domains. In terms of age differences, adolescents were more likely than young adults to use cannabis with greater frequency and specify cannabis use as their treatment target. While both adolescent and young adult cannabis users likely to exhibit symptoms of concurrent disorders across all three domains (i.e., internalizing, externalizing, and substance use disorders), substance use disorders was more likely to co-occur with externalizing disorders in adolescence and with internalizing disorders in young adulthood, although concurrence rates were high among all participants.

The findings of subgroup differences along the sex and age dimensions have implications for gender/sex- and developmentally informed research and care. From a sex/gender lens, the high needs observed among girls and young women are of particular note, particularly the three-way co-occurring problems in the internalizing, externalizing, and substance use domains. It may be that due to social norms, the mental health and substance use problems of girls and young women do not come to the attention of the healthcare sector until they are more advanced and complex, accounting for the smaller number of girls/young women in treatment, the high rates of complex three-way concurrent disorders, and the wider variety of substances used in the sample. In the adult literature, previous studies have shown that women seeking substance use treatment often have more severe needs than men, as they may face sex-specific barriers to treatment entry (for a review, see Tuchman, 2010). This merits further exploration, as there is a smaller evidence base available for substance use problems among women compared to men, particularly with regard to sex effects and treatment (Tuchman, 2010). Differences observed in substance use and mental health profiles between boys/young men and girls/young women and across age groups could indicate differences in needs, motivation for substance use, and help-seeking behaviors, while potentially suggesting differences in treatment strategies by subgroup. From a developmental perspective, our findings of higher rates of externalizing disorders in younger youth and increasing rates of internalizing disorders with older age are consistent with those of Chan, Dennis, and Funk (2008), including high rates of three-way concurrent disorders in young people. Together, these findings suggest that when screening for mental health and substance use problems, these sex and age-based variations in profiles could serve as a starting point to investigate further the clients' co-occurring problems and more quickly

identifying treatment strategies, leading to appropriate gender/sex- and developmentally informed care.

The large proportion of youth using cannabis on a daily basis and identifying cannabis as an important treatment target highlights the importance of providing cannabis-specific treatment options and pursuing research in this area to develop the most effective treatments, particularly in the context of the imminent legalization of non-medical cannabis. However, given the extremely high prevalence concurrent mental health problems among youth with substance use disorders, which has been well documented (e.g., Chan et al., 2008; Diamond et al., 2006) and was reflected in this study, concurrent disorder treatment should be a priority. Indeed, in a review of studies conducted in the U.S., 60% of youth with substance use disorders were found to have at least one co-occurring mental health disorder (Armstrong & Costello, 2002); our study found even higher rates of concurrent disorders, highlighting the importance of attending to overlapping needs in treatment settings. Best-practice policy papers have been published regarding the treatment of concurrent disorders (e.g., Health Canada, Canada's Drug Strategy, & Centre for Addiction and Mental Health, 2002; Watson, Carter, & Manion, 2014). Given the high rates of concurrent disorders among cannabis using youth, and the complex needs of these youth, the coordinated or integrated treatment of both categories of disorders is recommended (Watson et al., 2014).

However, the addictions and mental health service sectors have traditionally operated independently in distinct and often disjointed systems and struggled to coordinate services across the two sectors (Hawkins, 2009; Watson et al., 2014). In many jurisdictions the two separate sectors are financed by different funding agencies, adopt different conceptual models and underlying theories, and lack of consistencies in educational and licensing requirements (El-Guebaly, 2004; Hawkins, 2009; Watson et al., 2014). Concurrent disorder research, particularly research regarding three-way concurrent disorders marked by substance use disorders, internalizing disorders, and externalizing disorders, represents a significant research gap. This leaves a substantial proportion of service-seeking young people particularly vulnerable, especially young women with high rates of three-way concurrent disorders. Youth-specific, cannabis-specific treatment research is needed, with an emphasis on the co-occurring mental health challenges. Given the nearly universal use of cannabis among this sample of treatment seeking youth, the high rate of high-frequency cannabis use and concurrent disorders, and the substantial portion of young people with three-way concurrent disorders, it is essential that these research and service gaps be bridged. Further work is required to refine coordinated treatment modalities across the addictions and mental health service sectors, accounting for the nearly universal use of cannabis

among the targeted youth.

While some treatment recommendations are available for youth with concurrent disorders, in order to better serve youth with high rates of cannabis use and multiple intersecting needs as found in the current sample, a closer collaboration and integration among substance use and mental health treatment sectors is required. A model of integration with increasing national and international traction is the one-stop integrated care hub model, which brings together a variety of developmentally appropriate and youth-friendly services that address all aspects of the youth's health (Henderson et al., 2017; Hetrick et al., 2017; McGorry, Bates, & Birchwood, 2013). These models generally include services for addictions, mental health, and physical health, through a team of multidisciplinary care providers working in an integrative and collaborative way to serve youth with multiple intersecting needs. Capacity building among such integrated care hubs regarding the omnipresence of cannabis use and its important clinical implications for youth may strengthen such models and help them to best meet the needs of Canadian youth.

This study has a number of limitations. Firstly, the results of this study rely solely on self-reported data. This could affect the validity and reliability of the results. Secondly, the psychometric properties of the GAIN-SS extension items are unknown. The purpose of these extension items was to flag areas of concern for more in-depth assessment, rather than constituting psychometrically validated assessments; the results of the GAIN-SS extension items should therefore be interpreted with caution. Thirdly, only a subsample of participants was asked to report the age of onset for various substances, which may have affected the power to detect meaningful demographic differences. Finally, since this study reports on the profiles of youth accessing one specific treatment facility in a major metropolitan city in Canada, it needs to be replicated in other treatment contexts to examine the generalizability of the results.

5. Conclusion

This study of treatment-seeking youth highlights different profiles of use across substances, and the very high rates of concurrent disorders, including differences across sex/gender and age. Rates of cannabis use, combined with other substance use and three-way concurrent disorders, signal an urgent need to consider cannabis use in all aspects of substance use treatment and to continue developing treatment strategies that focus on two- and three-way concurrent disorders in sex/gender- and developmentally informed ways. Integrated service hub models for youth, which are designed to address multiple intersecting needs, may provide substantial benefits in this area. With the impending legalization of non-medical cannabis use in Canada, cannabis use rates and impacts among youth warrant close monitoring over the coming years.

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